Market Price Forecasting for Ugandan Staple Crops

A Multi-Model Time-Series Approach Using WFP Market Data and Climate Covariates

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Presentation Outline

1. Introduction & Problem Statement
2. Data Pipeline
3. Feature Engineering
4. Model Architectures
5. Results & Evaluation
6. Streamlit Dashboard
7. Challenges & Limitations
8. Conclusions & Future Work

The Problem: Food Insecurity through Price Volatility

- Agriculture = **24% of Uganda's GDP**, employs **70%+** of workforce
- Staple crops: **Maize, Beans, Cassava**
- Prices fluctuate **30–50%** between seasons
- Smallholder farmers sell post-harvest at **lowest prices**
- Existing WFP/FEWS NET systems are **retrospective**, not predictive

Impact

- ✓ Farmers optimise selling timing
- ✓ NGOs pre-position food aid
- ✓ Policymakers plan interventions

Our Solution

Build a machine learning system that forecasts crop prices **4–8 weeks ahead**

Research Question & Objectives

Research Question

How accurately can machine learning models forecast staple crop market prices 4–8 weeks ahead in Uganda, and which model – ARIMA, Prophet, or LSTM – achieves the best performance when rainfall is incorporated as a predictor?

Five Objectives:

1. **Collect & Merge** WFP prices + CHIRPS rainfall + World Bank commodity indices
2. **Engineer Features:** lags, rolling stats, seasonal encoding, climate covariates
3. **Implement 3 Models:** ARIMA, Facebook Prophet, LSTM
4. **Evaluate & Compare** using MAE, RMSE, MAPE (80/20 chronological split)
5. **Deploy** best model in interactive Streamlit dashboard

Three Heterogeneous Data Sources

Dataset	Source	Content	Role
WFP Food Prices	Humanitarian Data Exchange	Market prices (UGX) for crops across districts	**Target variable**
CHIRPS Rainfall	UC Santa Barbara satellite	Monthly rainfall (mm) by region	**Climate covariate**
World Bank Commodities	World Bank	Global crude oil & maize indices	**Global proxy**

Scope: 3 crops $\times$ 3 districts $\times$ 14 years (2010–2024)

Crops	Districts	Observations
Maize, Beans, Cassava flour	Kampala, Gulu, Mbale	$\sim$150–200 per pair

Data Integration Challenges

Challenge 1: Geographic Mismatch

- WFP: text names ("Gulu")
- CHIRPS: ISO P-codes ("UG1")
- **Solution:** Manual mapping dictionary

Challenge 2: Temporal Mismatch

- WFP: irregular weekly/monthly
- CHIRPS: dekadal (10-day)
- World Bank: "2023M06" format
- **Solution:** Resample all to Month-Start (MS)

Challenge 3: Missing Values

- Price gaps of 1–3 months
- **Solution:** Linear interpolation (limit=3)
- Covariates: forward-fill + backward-fill

Final Merged Dataset

Per (crop, district) pair:

- $\sim$150–180 monthly observations
- Columns: price, rainfall, oil, maize index
- All temporally aligned

Transforming Raw Data into AI-Ready Features

1. Price Lag Features

$$\text{price_lag}_k(t) = \text{price}(t-k)$$

$k \in \{1, 2, 4, 8\}$ – captures momentum & trends

2. Rolling Statistics

$$\text{roll_mean}_w(t) = \frac{1}{w}\sum_{i=0}^{w-1} \text{price}(t-i)$$

Windows: 4 months, 8 months
Also: rolling std. dev. (measures volatility)

3. Sinusoidal Seasonal Encoding

$$\text{month_sin} = \sin\left(\frac{2\pi m}{12}\right)$$

$$\text{month_cos} = \cos\left(\frac{2\pi m}{12}\right)$$

Preserves cyclical continuity (Dec $\leftrightarrow$ Jan)

4. Climate Lags

$$\text{rain_lag}_k(t) = \text{rainfall}(t-k)$$

$k \in \{1, 4, 8\}$ – rainfall today affects prices in 4 months (growing cycle)

5. Global Covariates: World Bank crude oil + maize indices (transport cost proxies)

Model 1: ARIMA – Classical Statistical Baseline

ARIMA(p, d, q):

$$y'_t = c + \sum_{i=1}^{p} \phi_i y'_{t-i} + \sum_{j=1}^{q} \theta_j \varepsilon_{t-j} + \varepsilon_t$$

- **AR(p):** Linear combination of past values
- **I(d):** Differencing for stationarity
- **MA(q):** Past forecast errors for self-correction

Order Selection: Exhaustive AIC grid search

$$\text{AIC} = 2k - 2\ln(\hat{L})$$

$p \in \{0..3\}$, $d \in \{0, 1\}$, $q \in \{0..3\} = 32$ candidates/pair

Strengths

- ✓ Interpretable coefficients
- ✓ No overfitting risk (few params)
- ✓ Works well on stationary data

Limitations

- × Linear assumptions
- × Univariate only
- × Struggles with regime changes

Model 2: Facebook Prophet – Additive Decomposition

Prophet Decomposition:

$$y(t) = g(t) + s(t) + \beta_{\text{rain}} \cdot \text{rainfall}(t) + \varepsilon_t$$

- $g(t)$ = **Piecewise linear trend** with automatic changepoint detection
- $s(t)$ = **Fourier seasonal component**:

$$s(t) = \sum_{n=1}^{N} \left(a_n \cos \frac{2\pi nt}{P} + b_n \sin \frac{2\pi nt}{P} \right)$$

- β_{rain} = Learned rainfall coefficient

Strengths

- ✓ Handles missing data
- ✓ Strong yearly seasonality
- ✓ External regressors

Limitations

- × Fixed Fourier seasonality
- × Spurious changepoints
- × Designed for business data

Model 3: LSTM – Deep Learning Neural Network

LSTM Gating Mechanism:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \qquad \text{Forget Gate}$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \qquad \text{Input Gate}$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \qquad \text{Output Gate}$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \qquad \text{Cell State}$$

Cell state = "memory highway" carries information across time steps without degradation

Solves **vanishing gradient problem** of standard RNNs

Our Architecture

Layer 1	LSTM (64 units)
Dropout	20%
Layer 2	LSTM (32 units)
Dropout	20%
Output	Dense (1 unit)
Window	6 months
Features	9 inputs
Optimiser	Adam
Loss	MSE
Epochs	100 (early stop)

LSTM: Multivariate Sliding Window Approach

9-Feature Input Vector (per time step)

`price`	Current price (UGX)
`rainfall`	Monthly rainfall (mm)
`wb_crude_oil`	Global oil index
`wb_maize`	Global maize index
`price_lag_1`	Price 1 month ago
`price_lag_2`	Price 2 months ago
`month_sin`	Seasonal indicator
`month_cos`	Seasonal indicator
`rain_lag_1`	Rainfall 1 month ago

Sliding Window

Examines **6 consecutive months** of all 9 features to predict **month 7's price**

Normalisation

MinMaxScaler: $x_{\text{scaled}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$

Ensures all features contribute equally

Evaluation Metrics

MAE

$$\frac{1}{n}\sum|y_i - \hat{y}_i|$$

Average error in UGX

Most interpretable

RMSE

$$\sqrt{\frac{1}{n}\sum(y_i - \hat{y}_i)^2}$$

Penalises large errors more

Sensitive to outliers

MAPE

$$\frac{100\%}{n}\sum\left|\frac{y_i - \hat{y}_i}{y_i}\right|$$

Scale-independent percentage

Cross-crop comparison

Evaluation Protocol: Strict chronological **80/20 split**

No shuffling $\rightarrow$ No data leakage $\rightarrow$ No future information in training

Aggregate Results: LSTM Dominates

Model	Avg MAE (UGX)	Avg RMSE	Avg MAPE
ARIMA	757.45	882.33	25.11%
Prophet	841.27	969.86	29.66%
LSTM	**503.13**	**606.40**	**17.65%**

LSTM Advantage

- **33% lower MAE** than ARIMA
- **40% lower MAE** than Prophet
- Best MAPE: **11.59%** (Gulu Maize)

Why LSTM Wins

1. Nonlinear modelling capacity
2. Multivariate (9 features vs 1)
3. Long-range memory via gating

District-Level Performance Breakdown

District	Crop	ARIMA MAPE	Prophet MAPE	LSTM MAPE
Kampala	Maize	40.66%	37.13%	**31.33%**
Kampala	Beans	48.35%	35.59%	**18.15%**
Kampala	Cassava flour	29.40%	28.40%	**18.84%**
Gulu	Maize	13.49%	26.39%	**11.59%**
Gulu	Beans	18.60%	16.18%	**12.22%**
Gulu	Cassava flour	22.67%	**12.34%**	20.01%
Mbale	Maize	**12.25%**	18.37%	14.22%
Mbale	Beans	**11.52%**	52.50%	11.78%
Mbale	Cassava flour	29.08%	40.07%	**20.75%**

Key Insight: LSTM wins **7 out of 9** pairs. ARIMA is competitive in Mbale (simpler, more stable market). Prophet wins only for Gulu Cassava (strong annual seasonality).

Interactive Web Dashboard

Dashboard Features:

1. **User Controls:** Select crop, district, forecast horizon
2. **Real-time LSTM Training:** Model trains live on-the-fly
3. **Interactive Plotly Chart:**
   - Historical prices (train split)
   - Test actuals vs LSTM predictions
   - Future forecast trajectory
4. **Performance Metrics:** MAE, RMSE, MAPE displayed
5. **Model Comparison Panel:** ARIMA vs Prophet vs LSTM side-by-side

Technology Stack

- **Framework:** Streamlit
- **Charts:** Plotly (interactive)
- **Backend:** TensorFlow/Keras
- **Caching:** `@st.cache_data`
- **Command:** `streamlit run app.py`

Live Demo

[SWITCH TO BROWSER]

`http://localhost:8501`

Challenges Faced & Overcome

1. Data Sparsity

Problem: Neural networks need 10,000+ rows; we had ∼200

Solution:

- Dropout (20%) – implicit ensemble
- Early stopping (patience=10)
- Compact architecture (64→32)

2. Geographic Mismatch

Problem: Three datasets use different spatial identifiers

Solution: Manual mapping dictionaries aligning districts to P-codes

3. Multi-step Forecasting

Problem: Error compounds recursively

Solution: Recursive strategy with updated lag features at each step

Known Limitations

- Single train/test split (vs. time-series CV)
- No confidence intervals
- ARIMA used without exogenous vars
- Models cannot predict black swan events (e.g., COVID)

Conclusions

What We Achieved

1. Built an **end-to-end ML pipeline**: data collection → preprocessing → feature engineering → modelling → evaluation → deployment
2. Merged **3 heterogeneous datasets** (WFP + CHIRPS + World Bank)
3. Trained and compared **3 fundamentally different models**
4. LSTM achieved **17.65% average MAPE** – competitive with published benchmarks
5. Deployed to an **interactive Streamlit dashboard** with model comparison

Future Work

- Transformer architectures
- Model ensembling (learned weights)
- Additional covariates (conflict, mobility)
- Probabilistic forecasts (intervals)
- Cloud deployment (AWS/GCP)
- SMS/USSD for farmer access

Thank You

Questions & Discussion

Repository: `https://github.com/Jamugisa-Peter-Paul/Price-forcasting`
Dashboard: `streamlit run app.py`

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